

## MODULE 3

### NANOTECHNOLOGY

Study of matter at atomic level is Nano technology . One nanometer(nm)  $10^{-9}$ (or one billionth) of a meter. Generally nanotechnology deals with structures of size 100 nm or smaller.

#### Nanofabrication

Nanofabrication involves 2 ways

- 'Top-down' fabrication: Starting with a chunk of materials and cutting it to nano scale.
- 'Bottom-up' fabrication: Starting with individual atoms and building up to nanostructure.

The theoretical predictions of the present Nanotechnology were made by Richard Feynman in 1959. In a talk entitled “there is plenty of room at the bottom” . Molecular self assembly is the science of automatically arranging molecules themselves into some useful structures through a “bottom up “ approach. The ability of one molecule to attract and bind to another is referred to as molecular recognition

#### Increase in surface to volume ratio

Nano material have a relatively larger surface area to volume ratio when compared to bulk solids. Consider a sphere of radius 'r' .

The surface area  $= 4\pi r^2$

Its volume  $= \frac{4}{3}\pi r^3$

$$\text{Ratio} = \frac{\text{surface area}}{\text{volume}} = \frac{3}{r}$$

Thus when the radius of the sphere decreases, its surface area to volume ratio increases. Hence as the particle size decreases a greater number of atoms are found at the surface compared to those inside. Thus nano material have more surface area per given volume compared with larger particles. It makes material more chemically reactive as chemical reaction occurs at surfaces.

When the size of the bodies decrease, physical properties of the bodies change. A number of physical properties(mechanical, electrical, optical ) change like this.

Examples of physical change at nanosize

- When nano scale is reached, the colour of gold changes, its melting point and chemical properties changes.

- Nano gold does not act like bulk gold
- Copper which is opaque become transparent.
- Stable materials like aluminium turn combustible.
- Solids turn into liquid at room temperature(gold)
- Insulators become conductors (silicon)
- Chemically inert materials like gold become good catalyst (reactive) at nano scales.

## **SIGINIFICANCE OF NANO SCALE**

The important factors that cause significant change in the properties of nano materials are

- Increased surface to volume ratio
- Quantum confinement effects

### **Quantum Confinement Effects**

When electron gets excited in a semi conductor due to thermal excitation, an electron-hole pair is created. The electron in the conduction band and hole in the valence band can be bound when they approach each other at a finite distance. This bound pair is called EXCITON. Bulk material have continuous energy levels. In the case of nano range, the energy level become discrete. As the size reduces energy level changes from continuous to discontinuous. This increases the band gap energy. Motion of the electron become restricted. the band gap and wavelength are inversely related to each other. Wavelength decrease with the decrease in size and proof is the emission of blue radiation

To understand this effect we break the word like quantum and confinement. The word confinement means to confine the motion of randomly moving electron to restrict its motion in specific energy levels. Quantum reflects the atomic domain of particles. The quantum confinement effects is observed when the size of the particle is too small to be comparable to the de-broglie wavelength of the electron

Quantum confinement is the restricted motion of randomly moving electron in specific energy levels when the dimensions of a material approach the de Broglie wavelength of electron

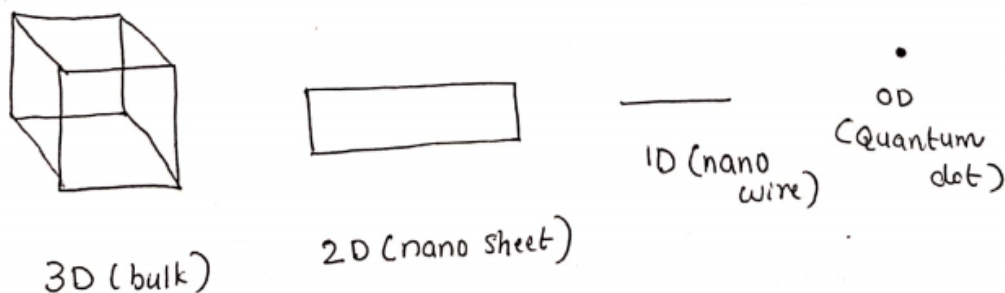
When the material size is sufficiently small(10nm or less) the electrons and holes are squeezed into a dimension that approaches a quantum measurement called “Exciton Bohr Radius”. When the particle size approaches the Bohr exciton radius, the quantum confinement effect increases .

### **Nanostructured materials**

The material which have at least one dimension in the order of nanometer are called nanostructured materials

Based on number of dimensions that are confined, nano structure is classified in to 4 categories

1. Zero dimensional (0D) nano structured material:- If all the three dimensions become nanometer range, we obtain a Zero dimensional nano structure called nano particles
2. One dimensional (1D) nano structured material:- If two dimensions are restricted and one dimension remains large, then the resulting structure is one dimensional nano structure called nano wires, nano rods and nano tube
3. Two dimensional (2D) nano structured material:- If one dimension is reduced to the nano range while the other two dimensions remain large, then a two dimensional nano structured material are obtained (nano sheet)
4. Three dimensional (3D) nano material:- all the 3 dimensions become large (bundles of nano wires)
  - Zero dimensional (0D) nano structured material - 3 D confinement
  - One dimensional (1D) nano structured material - 2 D confinement
  - Two dimensional (2D) nano structured material - 1 D confinement
  - Three dimensional (3D) nano material – 0 D confinement



#### Quantum sheets

Quantum sheets are two dimensional structure in which quantum confinement act only in one direction. Quantum confinement act along the thickness of the film. Electrons are free to move in xy plane of the film. So quantum confinement is in one dimension and are free to move in 2 dimensions. They are now used to make semiconductor lasers, LEDs and solar cell .

#### Quantum wires

A nano wire or quantum wire is a wire of dimensions of the order of nanometers. Quantum wire is a one dimensional structure. Quantum wires are formed when 2 dimensions of the system are confined ie by making a thin wire of semiconductor. Two direction have quantum confinement. It is only free to move along one dimension that is along the wire. A quantum wire is an electrically conducting wire in which quantum effects influence the transport properties. Such effects appear in the dimension of Nano meters. So they are called Nano wires. Electrical conductance of wire is found to be quantized in multiples of  $2e^2/h$ . quantum wires can be used as an electron wave guide. It is used to make high speed lasers.

#### Quantum dots

Quantum dots are zero dimensional structure in which the electron is confined in all three dimensions. Their energy states are quantized in all three directions. consider a tiny dot of semi conductor material then we have a 3D quantum confinement of charge carriers. That is there are no freely moving electrons or holes in any direction .This is called “quantum dot”. Although this quantum dot consists of thousands of atoms it behaves like a single atom with discrete energy levels rather than many atoms. When the size of the crystal become so small that it approaches the size of exciton bohr radius, then the electron energy levels will change from continuous to discrete values. That is there is a finite separation between energy levels. The energy gap increases with the decreases in the diameter of the dot. So quantum dot can emit any colour from blue to red depending on the size of the dot.

#### Variation of properties of nano materials

1. Physical properties
2. Chemical properties
3. Electrical properties
4. Optical properties
5. Magnetic properties
6. Mechanical properties

#### Electrical properties

- Ionization potential at nano sizes are higher than that at bulk solids.
- The change in electrical properties is not the same in all nano materials.
- In nano ceramics the electrical conductivity increases with the decrease in particle size.
- But in metals it decreases.
- In the case of bulk metals the energy level of electron is continuous where as in nano sized metals the energy levels are discrete.

#### Optical properties

- Depending on the particle size, same substance shows different colours.

- Gold spheres of 100 nm size appears orange while that of 50 nm size appears green

The particles can be made to emit or absorb specific wavelength of light by varying its size

#### **Mechanical properties**

- Nano crystalline nickel is as strong as hardened steel. Because of nano size many mechanical properties such as hardness, elastic modulus, fracture toughness, scratch resistance and fatigue strength are modified
- 30-50 % lower elastic moduli than conventional materials
- 2-7 times higher hardness
- Super plastic behaviour in brittle ceramics

#### **NANOCLUSTERS**

Their size is in between bulk materials and their atoms or molecules. Their properties are fundamentally different from those of discrete molecules and bulk solids

#### **Fullerenes**

They are spheroidal in structure. They are the largest molecules formed by the same atoms. A carbon molecule with chemical formula  $C_{60}$  containing 60 carbon atoms in the shape of a soccer ball. A mass number of 720 consisting of 60 carbon atoms, each of mass 12 was seen. Named after the architect buckminster fuller. Solid state: FCC structure.

#### **CARBON NANOTUBE**

When a single sheet of  $sp^2$  hybridized graphite (graphene) is curled in the form of a cylinder, carbon nanotubes are formed

### **Applications of Nanotechnology**

#### **Nanomedicine**

- Cell imaging
- Treating cancer
- For diagnostic purposes
- The drug consumption and side effects can be reduced significantly by targeted medicine.
- This is done depositing the medicine in the affected region only and in the correct dose
- NT helps to reproduce or repair damaged tissue.

## In Energy

Applications in the field of energy are

1. By reducing energy consumption  
by designing LEDs or Quantum Caged Atoms(QCAs) etc.
2. By increasing energy production efficiency  
by designing specific catalysts to improve the combustion efficiency in IC engines
3. By using more environmentally friendly energy systems  
such as fuel cells powered by hydrogen
4. By recycling batteries  
rechargeable batteries using nanomaterials

## chemistry and Environment

- Chemical catalysts from nanoparticles are highly useful
- Filtration : Nanofiltration helps
  1. Waste water management
  2. Air purification
  3. Energy storage devices

## Information and communication

- Memory storage devices such as Nano-RAM is developed using nanotubes
- Food
- There are more than thousand nanoproducts in the market.
- Eg. of food products are canola cooking oil, nanotea etc.

## Textiles

- Clothes made of nanofibers are stain-repellent and wrinkle-free
- These textiles can be washed less frequently

## Implication of Nanotechnology

- Nano Pollution
- Lack of Privacy
- Nano Weapons